An AI Ecosystem for Fake Job Scam Detection and Fraud-Aware Job Recommendation

INDIVIDUAL THESIS

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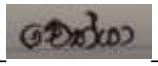
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ABSTRACT

Online recruitment platforms provide job seekers with convenient access to employment opportunities but also expose them to risks associated with fake job advertisements, fraudulent employers, and scam communications. Conventional job recommendation approaches primarily focus on relevance and may not adequately consider the potential risks associated with recommended opportunities. This research addresses the problem of incorporating multiple recruitment-fraud indicators into job recommendation and ranking while maintaining recommendation relevance. The aim is to develop a Risk-Aware Job Recommendation and Ranking component that balances job suitability and recruitment safety. The proposed approach uses Term Frequency–Inverse Document Frequency (TF-IDF) and Cosine Similarity to calculate skill-match scores between job seekers and available job opportunities. Fraud-risk indicators from fake-job detection, employer-legitimacy verification, and communication-scam detection are aggregated using the Entropy Weight Method (EWM). The resulting risk score is converted into a complementary safety score, and the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) is applied to rank jobs using relevance and safety as decision criteria. The component was evaluated using a public dataset containing 1,167 job records, 50 synthetic user profiles, and 516 ground-truth user-job records. Compared with a relevance-based baseline, the proposed approach maintained Precision@10 and MRR@10, while AP@10 and NDCG@10 showed small improvements. Average Risk@10 decreased from 0.1006 to 0.0900, Average Safety@10 increased from 0.8994 to 0.9100, and Unsafe Rate@10 decreased from 2.80% to 2.40%. These findings provide experimental evidence that incorporating fraud-risk information can improve the measured safety of recommendations without an observed reduction in relevance. The individual contribution is the design, implementation, and evaluation of the risk-aware recommendation and ranking component within the overall recruitment platform.

Keywords: Job Recommendation, Recruitment Fraud, Entropy Weight Method, TOPSIS, Risk-Aware Ranking

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AP@10	Average Precision at 10
DSR	Design Science Research
EWM	Entropy Weight Method
IDCG@10	Ideal Discounted Cumulative Gain at 10
MAP	Mean Average Precision
MRR@10	Mean Reciprocal Rank at 10
NDCG@10	Normalized Discounted Cumulative Gain at 10
NLP	Natural Language Processing
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
TF-IDF	Term Frequency–Inverse Document Frequency

1 INTRODUCTION

1.1 Overall project context

Online job platforms have made it easier for job seekers to find employment opportunities, but they have also created opportunities for recruitment fraud such as fake job advertisements, fraudulent employers, and scam communications. Existing job recommendation approaches mainly focus on matching users with relevant job opportunities based on their skills and preferences, without sufficiently considering whether the recommended opportunities are trustworthy. The overall research project therefore proposes an integrated framework that combines fraud detection and job recommendation to improve both the relevance and safety of online job recommendations.

1.2 Individual component context

This research focuses on the Risk-Aware Job Recommendation and Ranking component of the overall framework. The component uses TF-IDF and Cosine Similarity to determine the relevance between a user's skills and job descriptions. It then combines fraud risk indicators from fake job detection, employer legitimacy detection, and communication scam detection using the Entropy Weight Method, and applies TOPSIS to rank jobs based on both skill relevance and safety. The intended outcome is a ranked list of job opportunities that balances user suitability with the potential fraud risk associated with each job.

1.3 Problem statement

Existing job recommendation systems primarily focus on matching job seekers with suitable employment opportunities based on factors such as skills and job descriptions. However, a job with a high relevance score may still be associated with a fake job posting, an unreliable employer, or a scam communication. Therefore, there is a need for a recommendation approach that considers both job relevance and fraud risk when ranking employment opportunities.

1.4 Research gap

Existing research has separately addressed fraudulent job detection, content-based job recommendation, employment-risk prediction, and multi-criteria recommendation. However, the new literature review identified a lack of an integrated approach that combines multiple recruitment fraud indicators with job recommendation and ranking. In particular, existing approaches do not combine fake job detection, employer legitimacy, and communication scam risk with job relevance within a single recommendation process.

1.5 Aim and objectives

Aim

To develop a risk-aware job recommendation and ranking model that integrates skill matching and fraud risk analysis to recommend relevant and safer job opportunities.

Objectives

1. To develop a content-based job-matching approach using TF-IDF and Cosine Similarity to calculate the relevance of job opportunities to user profiles.
2. To aggregate fake-job, employer-legitimacy, and communication-scam risk indicators using the Entropy Weight Method and derive a corresponding safety score.
3. To develop a multi-criteria ranking approach using TOPSIS to combine job relevance and safety.
4. To evaluate the proposed approach against a relevance-based baseline using recommendation relevance and safety metrics.

1.6 Research questions or hypotheses

The main research question is “How can fraud risk be integrated into a job recommendation and ranking process while maintaining the relevance of recommended jobs?”

This question focuses on whether information from multiple fraud-detection sources can be incorporated into the recommendation process without making the recommendations substantially less relevant to the user's skills.

1.7 Scope limitations and assumptions

The scope of this research is limited to the Risk-Aware Job Recommendation and Ranking component. It covers skill matching using TF-IDF and Cosine Similarity, aggregation of three fraud-risk indicators using the Entropy Weight Method, safety-score calculation, and final ranking using TOPSIS. The component receives fraud-related outputs from the other research components rather than developing the underlying fake-job, employer-legitimacy, or communication-scam detection models itself.

The research is also limited by the quality and availability of the job and fraud-risk data used for evaluation. The proposed ranking depends on the risk indicators provided by the preceding components, meaning inaccuracies in those inputs can affect the final recommendations. The research focuses on the selected recommendation and ranking techniques and does not claim to detect every possible form of recruitment fraud.

1.8 Individual contribution

The individual contribution of this research is the design and development of the Risk-Aware Job Recommendation and Ranking component within the overall four-component framework. The component performs skill matching using TF-IDF and Cosine Similarity, aggregates the three fraud-risk indicators using the Entropy Weight Method, converts the resulting risk into a safety score, and applies TOPSIS to produce the final ranked job recommendations.

The component therefore provides the integration point between the fraud-detection outputs produced by the other components and the final job recommendation process. Its purpose is to incorporate both job relevance and safety into the ranking rather than relying on skill relevance alone.

1.9 Thesis structure

This thesis is organized into nine chapters.

Chapter 1 introduces the research context, problem, research gap, objectives, research questions, scope, and individual contribution.

Chapter 2 presents the literature review and related work, including the updated literature review conducted for this research.

Chapter 3 describes the research methodology, including the research design, data, procedures, and evaluation methods.

Chapter 4 presents the proposed component design and architecture.

Chapter 5 describes its implementation and integration with the other components. Chapter 6 presents the results and validation.

Chapter 7 discusses the findings, contributions, and limitations.

Chapter 8 discusses commercialisation and sustainability aspects.

Chapter 9 concludes the research and presents possible future directions. This structure follows the Individual Thesis template provided for the research project.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This literature review focuses on research relevant to risk-aware job recommendation and ranking, with particular attention to four areas: fraudulent job recruitment detection, content-based job recommendation, risk-informed job recommendation, and multi-criteria recommendation and ranking. The review was updated after the initial research proposal to reflect more recent and directly relevant studies. Four studies were selected for detailed comparison based on their relevance to the proposed Risk-Aware Job Recommendation and Ranking component. These include studies on fraudulent job recruitment detection, TF-IDF-based job recommendation, layoff-risk-aware job recommendation, and a hybrid recommendation approach using Entropy Weighting and TOPSIS.

The selected studies were examined according to the problem addressed, techniques used, type of risk or recommendation information considered, and limitations relevant to the proposed research. The review therefore does not treat the studies as isolated solutions. Instead, their approaches are compared to understand which aspects of job matching, risk assessment, and ranking have already been addressed and where these areas remain disconnected. The review is specifically scoped to the requirements of this research component and does not attempt to provide an exhaustive review of all research on recruitment fraud or recommender systems.

The reviewed studies provide four relevant perspectives. Chiraratanasopha and Chay-intr (2022) address fraudulent job recruitment using fraud-oriented features and machine learning classifiers, but their work focuses on fraud detection rather than job recommendation. Winardi et al. (2025) demonstrate the use of TF-IDF and Cosine Similarity for content-based job recommendation, but the recommendation process does not incorporate recruitment fraud or employer credibility. Honmane et al. (2026) combine employment-risk prediction with job recommendation, although the risk considered is employee layoff probability rather than recruitment fraud. Canay (2026) investigates a hybrid recommendation approach incorporating Entropy Weighting and TOPSIS for multi-criteria ranking, but does not consider recruitment fraud or job safety.

This comparison provides the basis for the subsequent sections of this chapter. The literature is first examined in terms of the domain and theoretical foundations underlying job recommendation, similarity measurement, risk aggregation, and multi-criteria ranking. The relevant studies are then analysed in greater detail before existing techniques are compared with the approach proposed in this research. Finally, the findings are synthesised to establish the research gap and justify the conceptual framework of the Risk-Aware Job Recommendation and Ranking component.

2.2 Domain and theoretical foundations

Job Recommendation Systems

Job recommendation systems are designed to help job seekers identify employment opportunities that are relevant to their profiles. A recommendation process generally requires some representation of the user's characteristics and the available job opportunities, followed by a method for determining the suitability of each job. In the context of job recommendation, relevant information may include skills, education, experience, interests, salary preferences, and location. The reviewed work by Winardi et al. (2025), for example, uses user profile and job information including salary, education, skills, interests, and location to generate recommendations through content-based filtering.

Job recommendation approaches can use different forms of filtering and matching. A recent review of job recommender systems identifies content-based, collaborative, hybrid, and knowledge-based approaches among the commonly studied approaches, with similarity-based techniques such as Cosine Similarity and TOPSIS also being used in job recommendation research. For the present research, a content-based approach is appropriate because the proposed system needs to determine how closely a user's skills correspond to the requirements contained in a job description.

However, relevance alone does not necessarily indicate that a job opportunity is suitable for recommendation. A job may have a strong skill match while also having a high level of recruitment-related risk. Therefore, the proposed research extends the recommendation process by treating skill relevance and job safety as separate decision criteria rather than using skill similarity as the only basis for ranking.

Content-Based Job Matching

Job recommendation approaches can use different forms of filtering and matching. A recent review of job recommender systems identifies content-based, collaborative, hybrid, and knowledge-based approaches among the commonly studied approaches, with similarity-based techniques such as Cosine Similarity and TOPSIS also being used in job recommendation research. For the present research, a content-based approach is appropriate because the proposed system needs to determine how closely a user's skills correspond to the requirements contained in a job description.

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The advantage of content-based matching for this research is that it can generate a relevance score directly from the information available for an individual user and job. It does not require previous interactions from similar users, which is useful when the objective is to evaluate the relationship between a user's skills and a specific job posting.

Nevertheless, content-based matching by itself does not represent whether the job is trustworthy. This limitation is important to the current research because the literature reviewed shows that the TF-IDF-based recommendation approach focuses on profile and job similarity and does not incorporate fraudulent job postings, employer credibility, or recruitment scams.

TF-IDF and Cosine Similarity

TF-IDF

Term Frequency–Inverse Document Frequency (TF-IDF) is a text representation technique used to assign weights to terms according to their importance within a collection of documents. In the proposed research, TF-IDF is used to convert the textual information from the user profile and job descriptions into numerical vectors that can be compared.

The basic idea is that a term that occurs frequently within a document but is less common across the document collection can provide more useful information for distinguishing that document. Conversely, terms that occur across many documents provide less distinguishing information. This makes TF-IDF suitable for representing job-related terms where some skills or words may be more informative than common terms.

The use of TF-IDF for job recommendation is supported by the reviewed literature. Winardi et al. (2025) use TF-IDF as part of a content-based job recommendation system, while other job recommendation research has also used TF-IDF together with Cosine Similarity to compare job-related text.

For this research, TF-IDF is therefore used as the text representation stage, rather than as the final ranking method. Its output provides the numerical representation required for the subsequent similarity calculation.

Cosine Similarity

Once the user profile and job descriptions have been represented as TF-IDF vectors, Cosine Similarity can be used to measure their similarity. The measure compares the angle between two vectors rather than directly comparing their absolute magnitudes. For two vectors *A* and *B*, the similarity can be represented as:

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{||A||\,||B||}$$

where:

- A = TF-IDF vector representing the user profile
- B = TF-IDF vector representing a job description
- $A \cdot B$ = dot product of the two vectors
- $||A||$ and $||B||$ = magnitudes of the two vectors

A higher similarity value indicates that the two text representations are more closely aligned. In the proposed system, this value is used as the Skill Match Score for each candidate job. Cosine Similarity is also identified as a commonly used similarity technique in job recommender research.

It is important to distinguish this score from the final recommendation score. Cosine Similarity measures relevance between the user and the job; it does not measure job safety. This distinction is central to the proposed risk-aware approach.

Recruitment fraud introduces an additional dimension that is not captured by conventional relevance-based recommendation. A job posting may contain appropriate skills and responsibilities and therefore achieve a high similarity score while still presenting risks to the job seeker.

Chiraratanasopha and Chay-intr (2022) demonstrate that fraudulent job advertisements can be analysed using features designed to represent characteristics associated with fraudulent recruitment, including missing information, exaggeration, and credibility. Their study focuses on classifying job advertisements as legitimate or fraudulent and reports strong classification performance for its best model.

However, fraud detection and job recommendation address different stages of the decision process. Fraud detection determines whether evidence associated with a job indicates potential fraudulent behaviour, whereas recommendation determines which available jobs should be presented more prominently to a particular user. The comparison of the selected literature indicates that these functions have generally been addressed separately.

The proposed research therefore treats fraud information as an additional decision factor rather than replacing the relevance calculation. Three risk indicators are considered:

1) Fake job risk
2) Employer legitimacy risk
3) Communication scam risk

These indicators are supplied by the corresponding components of the overall four-component system. The Risk-Aware Job Recommendation and Ranking component does not develop those detection models itself; instead, it integrates their outputs into the recommendation process. The overall system architecture describes these three components as independently producing risk scores that are subsequently consumed by the recommendation component.

Entropy Weight Method for Risk Aggregation

When several risk indicators are available, simply assigning the same importance to every indicator may not reflect the information contained in the data. The proposed research therefore uses the Entropy Weight Method (EWM) to determine the relative weight of the three fraud-risk indicators.

The basic purpose of EWM is to derive criterion weights according to the amount of information or variation represented by each criterion. In this research, the three criteria correspond to the fake-job, employer-legitimacy, and communication-scam risk indicators.

The use of entropy-based weighting is relevant to the proposed system because the three indicators originate from different fraud-detection components and may not contribute equally to the final risk assessment. Rather than defining their weights entirely through subjective assumptions, EWM provides a data-driven mechanism for determining their relative contribution.

The research proposal specifies the risk aggregation process as a weighted combination of the three normalized risk indicators:

$$R = w_{1R_f} + w_{2R_e} + w_{3R_c}$$

where:

- R = aggregated fraud risk score
- R_f = fake-job risk
- R_e = employer legitimacy risk
- R_c = communication scam risk
- w_1, w_2, w_3 = weights obtained using EWM.

The resulting risk score is represented on a scale from 0 to 1, where a higher value indicates greater fraud risk

This approach is also consistent with the selected Canay (2026) study, which combines entropy-based weighting with TOPSIS in a multi-criteria recommendation framework. However, the criteria and purpose differ: that study applies the approach to skills-aware talent recommendation, whereas the present research applies entropy weighting

specifically to multiple recruitment-fraud indicators before incorporating the resulting safety measure into job ranking.

Safety Score

The aggregated risk score needs to be represented in a form that can be used alongside the skill match score during recommendation. The proposed model therefore converts the fraud risk into a complementary Safety Score:

$$Safety\ Score = 1 - Risk\ Score$$

A higher Safety Score represents lower aggregated fraud risk, while a lower Safety Score represents greater risk. This transformation allows the recommendation stage to treat both criteria as benefit criteria, where a higher value is preferred.

For example, a job with a high skill match but high fraud risk may have a strong relevance score but a low safety score. Conversely, a job with slightly lower skill relevance but substantially better safety may become preferable during multi-criteria ranking.

This is important because the objective of the proposed model is not simply to remove risky jobs or recommend the safest jobs regardless of relevance. Instead, the model attempts to balance the two dimensions: the job should be relevant to the user's skills while also presenting a lower level of recruitment-related risk.

TOPSIS for Multi-Criteria Job Ranking

Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) is a multi-criteria decision-making technique used to rank alternatives according to their distance from an ideal solution and an anti-ideal solution.

TOPSIS is appropriate for the proposed research because each candidate job needs to be evaluated using more than one criterion. After the preceding stages, every candidate job has at least two values relevant to the final decision:

- Skill Match Score — higher values are preferred.
- Safety Score — higher values are preferred.

The general principle of TOPSIS is to identify an ideal alternative that has the most desirable values for the selected criteria and an anti-ideal alternative representing the least desirable values. Each candidate is then evaluated according to its relative closeness to these two reference points. A candidate that is closer to the ideal solution and farther from the anti-ideal solution receives a more favourable ranking.

The use of TOPSIS is supported by the selected literature. Canay (2026) combines Entropy Weighting and TOPSIS within an interpretable multi-criteria recommendation framework. The broader review of job recommender systems also identifies TOPSIS among similarity-based and multi-criteria techniques used in job recommendation research.

In the proposed research, TOPSIS is therefore not used to calculate skill similarity or fraud risk independently. Instead, it serves as the final decision-making mechanism that combines the outputs of the previous stages. The resulting ranking reflects both the relevance of the job to the user and the safety of the opportunity.

Theoretical Basis of the Proposed Approach

The proposed Risk-Aware Job Recommendation and Ranking component is based on the combination of three main approaches: content-based skill matching, fraud-risk aggregation, and multi-criteria ranking. Each method performs a different function within the recommendation process. TF-IDF and Cosine Similarity are used to measure the relevance between the user's skills and available jobs, the Entropy Weight Method is used to aggregate the fraud-risk indicators, and TOPSIS is used to produce the final ranking based on both relevance and safety. This separation allows the recommendation process to consider job suitability and fraud risk as distinct but related criteria. The overall methodology is derived from the technical approach defined for the proposed component.

The first stage is skill matching. The user's profile and the job descriptions are converted into TF-IDF representations, after which Cosine Similarity is used to measure the similarity between them. The resulting value represents the Skill Match Score, where a higher value indicates stronger alignment between the user's skills and the requirements of a job. This follows the proposed methodology, where TF-IDF is

used to represent textual information and Cosine Similarity is used to calculate the similarity between the user profile and job descriptions.

User Profile + Job Description → TF-IDF → Cosine Similarity → Skill Match Score

The second stage incorporates the three fraud-related indicators received from the other components of the overall system. These indicators represent fake-job risk, employer legitimacy risk, and communication scam risk. The Entropy Weight Method is used to determine the relative contribution of these indicators and aggregate them into an overall Risk Score. The risk score is then converted into a Safety Score, where a higher safety score represents lower aggregated fraud risk.

Fraud Indicators → EWM → Risk Score → Safety Score

The final stage combines the Skill Match Score and Safety Score as the two criteria used for recommendation ranking. TOPSIS is applied as the multi-criteria decision-making method to identify job alternatives that are closest to the ideal combination of high relevance and high safety. Therefore, the final ranking is not based solely on skill similarity but considers both the suitability and safety of each job opportunity.

Skill Match + Safety → TOPSIS → Risk-Aware Ranking

Overall, the proposed approach follows the sequence of matching jobs to the user's skills, aggregating fraud-related risks, converting risk into safety, and ranking jobs using both relevance and safety. This provides the theoretical basis for generating recommendations that aim to balance job suitability with protection against recruitment-related fraud. The individual techniques are supported by the methodology of the proposed system, while their combination forms the basis of the Risk-Aware Job Recommendation and Ranking component.

2.3 Related research

Research related to the proposed Risk-Aware Job Recommendation and Ranking component can be grouped into four main areas: fraudulent job detection, content-based job recommendation, risk-informed career recommendation, and multi-criteria recommendation. The following studies were selected because each contributes to one or more aspects of the proposed approach.

Fraudulent Job Recruitment Detection

Chiraratanasopha and Chay-intr [1] investigated the detection of fraudulent job advertisements by developing features based on characteristics associated with fraudulent recruitment. Instead of relying only on conventional term-frequency information, the study introduced features related to missing information, exaggeration, and credibility. These features were generated from the EMSCAD dataset and used with machine learning classifiers to distinguish fraudulent and legitimate job advertisements. The best reported model achieved an accuracy of 97.64%, with a precision of 0.97 and recall of 0.99.

This study is particularly relevant to the proposed research because it demonstrates that fraud-related information can be extracted from job advertisements and represented as measurable indicators. However, its main objective is the classification of fraudulent job advertisements. The study does not extend the detected fraud information into a personalized job recommendation or ranking process. Therefore, while it provides a basis for the fraud-risk aspect of the proposed component, an additional mechanism is required to use such information when selecting and ranking jobs.

Content-Based Job Recommendation Using TF-IDF

Winardi et al. [2] proposed a content-based job recommendation system using TF-IDF and Cosine Similarity. The system analyses job descriptions together with job-seeker profiles and generates recommendations based on attributes including salary, education, skills, interests, and job location. The authors implemented the approach using Scikit-learn and reported a Mean Average Precision (MAP) score of 0.798.

The study is directly relevant to the skill-matching stage of the proposed component. In particular, the use of TF-IDF and Cosine Similarity provides a suitable approach for representing textual job information and measuring the similarity between a job seeker's profile and available jobs. However, the recommendation process is primarily concerned with matching job characteristics with user preferences and qualifications. Fraud risk, employer legitimacy, and recruitment safety are not included as

recommendation criteria. Therefore, the approach provides the relevance component of the proposed model but does not address the safety dimension.

Risk-Aware Job Recommendation

Honmane et al. [3] proposed an integrated machine learning framework that combines job-risk assessment with career recommendation. The framework predicts an individual's risk of job loss using information such as employment history, skills, and industry trends. Random Forest, XGBoost, and Logistic Regression are used for risk classification, after which the system provides relevant job recommendations and career-transition support. The study therefore demonstrates how risk assessment can be incorporated into a recommendation-oriented system rather than treating risk prediction as an independent task.

However, the risk considered in this study is the individual's risk of job loss, rather than the risk associated with the job opportunity itself. This distinction is important for the proposed research. The proposed component is concerned with recruitment-related risks, including fake job postings, employer legitimacy, and communication scams. Therefore, although Honmane et al. demonstrate the value of combining risk information with job recommendation, their risk dimension differs from the fraud-risk dimension addressed in this research.

Multi-Criteria Recommendation Using Entropy Weighting and TOPSIS

Canay [4] proposed an interpretable hybrid recommendation framework called CF-RL-TOPSIS, which combines collaborative filtering, a reinforcement-style branch, and an entropy-weighted TOPSIS branch. The approach is designed for skills-aware talent recommendation and considers multiple occupation-related criteria when producing rankings. The study evaluates the approach using the JobHop and Karrierewege talent-history benchmarks and examines its ranking performance using metrics including NDCG@5 and paired statistical tests.

This study is relevant to the proposed research because it demonstrates the use of entropy-based weighting together with TOPSIS within a recommendation framework. These techniques provide a basis for handling multiple criteria and producing an interpretable ranking. However, the criteria considered in Canay's framework relate to

talent and career progression rather than recruitment fraud or job safety. In addition, the work is available as a 2026 preprint rather than being treated here as an established journal publication.

Comparative Analysis of Related Research

The reviewed studies demonstrate that important parts of the proposed research problem have already been investigated independently. Chiraratanasopha and Chay-intr provide an approach for identifying fraudulent job advertisements, while Winardi et al. demonstrate content-based job matching using TF-IDF and Cosine Similarity. Honmane et al. show that risk information can be incorporated into a recommendation-oriented system, although their risk relates to potential job loss. Canay demonstrates the application of entropy weighting and TOPSIS for multi-criteria recommendation and ranking.

The relationship between these studies and the proposed component is therefore complementary. TF-IDF and Cosine Similarity provide a basis for measuring job relevance, while fraud detection research provides the basis for obtaining recruitment-risk information. Risk-aware recommendation demonstrates the value of incorporating risk into recommendation, and entropy weighting with TOPSIS provides a mechanism for combining multiple criteria during ranking. However, the reviewed studies address these aspects separately rather than within the specific fraud-aware job recommendation process proposed in this research.

2.4 Existing systems products and techniques

Content-Based Job Recommendation Systems

Content-based job recommendation systems primarily use information contained in user profiles and job postings to identify suitable opportunities. A common approach is to represent textual information using TF-IDF and calculate Cosine Similarity between user and job representations. Winardi et al. used this approach to generate job recommendations based on attributes including skills, education, salary, interests, and location, demonstrating its suitability for matching job characteristics with user information.

More broadly, content-based job recommendation research has used techniques such as TF-IDF, Cosine Similarity, Word2Vec, and other natural language processing methods to represent and compare job-related information. A recent systematic review of job recommender systems found that content-based approaches remain an important category within job recommendation research, with similarity-based and NLP techniques frequently used for matching.

The main advantage of this approach for the proposed research is its direct relationship between user skills and job requirements. However, content similarity alone provides information about relevance rather than whether a job opportunity is trustworthy. Therefore, it is suitable as the relevance component but requires an additional safety-related criterion for the proposed system.

Fraud Detection Techniques for Recruitment

Fraud detection systems focus on identifying characteristics that distinguish legitimate recruitment opportunities from fraudulent ones. Chiraratanasopha and Chay-intr used fraud-oriented features related to missing information, exaggeration, and credibility and applied machine learning classification to fraudulent job advertisements. Their results demonstrate that recruitment fraud can be represented through measurable indicators rather than relying only on general textual similarity.

However, a fraud detection system normally produces a classification or fraud-related assessment rather than a complete recommendation ranking. For the proposed component, fraud-related outputs from the other components of the overall framework are therefore treated as risk indicators that can be incorporated into the subsequent recommendation process. This allows fraud information to influence the ranking rather than stopping at fraud classification.

Risk-Aware Recommendation Techniques

Risk-aware recommendation approaches extend conventional recommendation by considering an additional risk dimension. Honmane et al. developed a system that combines risk prediction with job recommendation, using Random Forest, XGBoost, and Logistic Regression to classify an individual's risk of job loss before providing career recommendations.

This approach demonstrates that recommendation can consider information beyond simple job relevance. However, the risk dimension is different from the one considered in this research. The system focuses on employment and layoff risk, whereas the proposed component considers recruitment fraud risk associated with the job opportunity. Consequently, the proposed approach requires fraud indicators that specifically reflect fake jobs, employer legitimacy, and scam communication.

Multi-Criteria Ranking Techniques

When recommendations need to consider multiple criteria simultaneously, multi-criteria decision-making methods can be used to produce a final ranking. TOPSIS is one such technique and ranks alternatives according to their relative closeness to an ideal solution. Canay's work demonstrates the use of entropy-based weighting together with TOPSIS in a skills-aware recommendation framework, showing how multiple criteria can be incorporated into an interpretable ranking process.

For the proposed component, TOPSIS is used after the relevance and safety values have been calculated. The two criteria are Skill Match Score and Safety Score, allowing a job to be evaluated based on both its suitability for the user and its associated safety level. The Entropy Weight Method is applied earlier in the process to determine the contribution of the different fraud-risk indicators.

Comparison with the Proposed Approach

The existing approaches provide the main technical foundations for the proposed component, but they address different parts of the problem. Content-based systems provide methods for measuring job relevance, fraud detection systems provide methods for identifying recruitment-related risk, risk-aware recommendation demonstrates the value of incorporating risk into recommendation, and multi-criteria methods provide mechanisms for combining several evaluation criteria.

The proposed approach brings these techniques together in a single recommendation pipeline. TF-IDF and Cosine Similarity are used to calculate job relevance, the Entropy Weight Method aggregates the three fraud-risk indicators into an overall risk score, the risk score is converted into a safety score, and TOPSIS produces the final ranking based on relevance and safety. This combination is the main technical distinction

between the proposed component and the individual approaches reviewed in the literature. The reviewed studies provide support for the individual techniques, while their integration is used in this research to address the specific problem of fraud-aware job recommendation.

2.5 Gap synthesis and conceptual framework

Gap Synthesis

The comparison of the selected studies identifies several limitations relevant to the proposed research. Chiraratanasopha and Chay-intr focus on detecting fraudulent job advertisements, but the detection output is not incorporated into a subsequent recommendation or ranking process. Winardi et al. demonstrate the use of TF-IDF and Cosine Similarity for content-based job recommendation, but the recommendation process does not account for recruitment fraud risk. Honmane et al. integrate risk assessment with job recommendation, but the risk considered is related to employee layoff probability rather than the risk of fraudulent recruitment. Canay demonstrates the use of Entropy Weighting and TOPSIS within a multi-criteria recommendation framework, but the approach is not designed around recruitment fraud or job safety.

Therefore, the reviewed literature indicates a gap at the integration level. The individual techniques required for the proposed component already exist, but their combination for recruitment fraud-aware job recommendation has not been addressed within the selected literature. In particular, there is a need to connect multiple recruitment fraud indicators with job relevance and incorporate both aspects into the final ranking process.

The proposed research addresses this gap by integrating three stages: skill matching, fraud-risk aggregation, and multi-criteria ranking. TF-IDF and Cosine Similarity are used to calculate job relevance, the Entropy Weight Method is used to combine fake-job, employer-legitimacy, and communication-scam risk indicators, and TOPSIS is used to rank jobs using both skill relevance and safety. This approach follows the methodology defined for the proposed component.

Conceptual Framework

The conceptual framework of the proposed Risk-Aware Job Recommendation and Ranking component consists of three sequential stages.

- **Stage 1 – Skill Matching:**

  User profile information and job description data are processed using TF-IDF and Cosine Similarity. This produces a Skill Match Score representing the relevance of each job to the user's skills.

- **Stage 2 – Risk Aggregation:**

  Three fraud-related indicators are received from the other components of the overall system: fake-job risk, employer legitimacy risk, and communication scam risk. The Entropy Weight Method determines the contribution of these indicators and combines them into an overall Risk Score. The risk score is then converted into a Safety Score using:

  Safety Score = 1 − Risk Score

- **Stage 3 – Risk-Aware Ranking:**

  The Skill Match Score and Safety Score are provided to the TOPSIS ranking method. TOPSIS evaluates the candidate jobs according to both criteria and produces the final Risk-Aware Ranked Job List.

This framework positions the proposed component as the integration point between the fraud-detection modules and the final recommendation process. The three preceding components provide independent fraud-risk information, while the proposed component combines these signals with job relevance to produce the final ranking. The overall architecture therefore separates fraud detection from recommendation while allowing their outputs to contribute to a unified decision-making process.

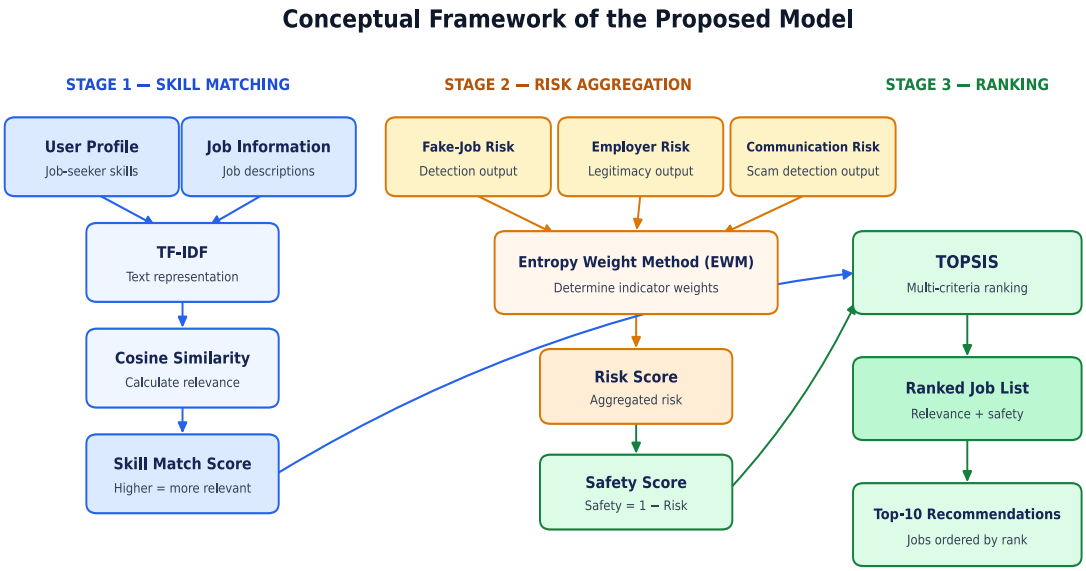


Figure 1: Conceptual framework of the Model

3 RESEARCH METHODOLOGY

3.1 Research design

This research follows a Design Science Research (DSR) approach because its primary purpose is to design, develop, and evaluate a computational artifact that addresses the identified problem of incorporating recruitment fraud risk into job recommendation and ranking. The research therefore involves both the development of a functional recommendation component and the empirical evaluation of its performance.

The artifact developed in this research is the Risk-Aware Job Recommendation and Ranking component within the overall four-component recruitment fraud detection and recommendation framework. The component integrates job relevance with fraud-related safety information during the recommendation process. It uses TF-IDF and Cosine Similarity to determine the relevance between user profiles and job postings, the Entropy Weight Method to aggregate fraud-risk indicators from the preceding components, and TOPSIS to rank candidate jobs using relevance and safety as decision criteria. The research proposal defines these three stages as the core technical approach of the component.

The research follows an artifact development and quantitative evaluation process. First, the recommendation problem and relevant research gap were identified through the literature review. The findings were then used to establish the conceptual framework and design the proposed recommendation and ranking approach. The component was subsequently implemented and integrated with the outputs of the fake-job detection, employer-legitimacy verification, and communication-scam detection components. These preceding components provide the fraud-related risk indicators used by the proposed ranking process rather than being developed within this individual component.

The developed component was evaluated experimentally using synthetic test-user profiles and a public job-posting dataset. A conventional relevance-based recommendation approach using TF-IDF and Cosine Similarity was used as the baseline, while the proposed approach additionally incorporated the fraud-risk information and applied the Entropy Weight Method and TOPSIS. The two approaches

were evaluated using the same test users and job data so that the effect of incorporating fraud risk into the recommendation process could be examined.

The evaluation considers two main aspects of the research question: recommendation relevance and recommendation safety. Relevance measures determine whether the recommended jobs correspond to the prepared ground-truth relevance information, while safety measures examine the fraud-risk characteristics of the recommended jobs. The results of the baseline and proposed approaches are subsequently compared to determine whether fraud-risk information can be incorporated into the ranking process while maintaining recommendation relevance.

Overall, the research follows the sequence of problem identification, literature review, conceptualisation, artifact design, implementation, integration, baseline comparison, and quantitative evaluation. This design connects the development of the technical artifact with empirical evidence required to assess the research question.

3.2 Data participants or study materials

This research does not involve human participants or the collection of personal data from real job seekers. Instead, the evaluation uses a publicly available job-posting dataset, synthetic test-user profiles, a ground-truth relevance dataset, and fraud-risk information generated by the other components of the overall research framework. The use of synthetic profiles allows the recommendation component to be evaluated under different job-seeker scenarios without requiring personal information from real individuals.

The primary study material is a public Kaggle job-posting dataset containing 1,167 job records. The dataset provides job-related information used for skill matching and recommendation generation, including job titles, job descriptions, and skill-related information. The job records represent the employment opportunities available to the recommendation component during the evaluation.

Synthetic test-user profiles were used to represent different job-seeker scenarios. The evaluation consisted of 50 test-user profiles distributed across five job domains: Information Technology, Business Development, Finance, Sales, and Human Resources, with 10 profiles representing each domain. The profiles were generated

using skills derived from selected job records and were used as inputs to the skill-matching process. The evaluation therefore represents simulated users rather than actual participants.

In addition to the job and test-user data, the proposed approach uses three fraud-related risk indicators produced by the preceding components of the overall research framework: fake-job risk, employer-legitimacy risk, and communication-scam risk. These indicators are inputs to the risk aggregation stage of the proposed component. They are not independently generated by the individual recommendation component.

A ground-truth dataset containing 516 user-job records was used to assess recommendation relevance. The ground-truth records associate test users with job postings and contain relevance information used for evaluating the generated recommendation lists. The baseline and proposed approaches were evaluated using the same test-user profiles and job-posting dataset to maintain a consistent comparison.

For each of the 50 test users, both the baseline and proposed approaches generated a Top-10 recommendation list, resulting in 500 recommendation records for each approach. The baseline generated recommendations using skill matching based on TF-IDF and Cosine Similarity, while the proposed approach additionally incorporated the fraud-risk indicators, calculated risk and safety scores, and applied TOPSIS to produce the final ranking. The evaluation configuration and data therefore allow the proposed approach to be compared against a relevance-based recommendation approach under the same test conditions.

3.3 Experimental or study procedure

The experimental procedure was conducted in a sequence of data preparation, recommendation generation, risk-aware ranking, and evaluation stages. First, the public job-posting dataset and the synthetic test-user profiles were prepared for the experiment. The job records provided the available employment opportunities, while each test-user profile provided the skills used to calculate job relevance.

The first recommendation stage performed skill matching between each test-user profile and the available job postings. The user's skills and job-related text were represented using TF-IDF, and Cosine Similarity was used to calculate a skill-match

score for each job. The resulting jobs were ordered according to their skill-match scores.

Two recommendation approaches were then generated using the same test users and job dataset. The baseline approach used the skill-match scores obtained from TF-IDF and Cosine Similarity and ranked the jobs according to relevance. The baseline implementation selected the top 10 jobs for each test user.

The proposed approach followed the same skill-matching stage but additionally incorporated the fraud-risk information supplied by the preceding components. The three risk indicators were processed using the Entropy Weight Method to obtain the relative weights of the fake-job, employer-legitimacy, and communication-scam risk criteria. These weighted indicators were then aggregated into an overall risk score for each job. The risk score was converted into a safety score using the complementary safety transformation defined for the proposed model. The resulting Skill Match Score and Safety Score were then supplied to the TOPSIS ranking process.

After TOPSIS ranking, the proposed approach selected the top 10 jobs for each test user. The recommendation-generation process therefore produced two Top-10 lists for every test user: one from the relevance-based baseline and one from the proposed risk-aware approach. Both sets of recommendations were stored for subsequent evaluation.

The recommendation lists were then evaluated using the prepared ground-truth relevance information and the fraud-risk information associated with the recommended jobs. Relevance evaluation examined the quality of the ranked recommendations using ranking and retrieval measures, while safety evaluation examined the aggregated risk and safety characteristics of the Top-10 recommendations. The baseline and proposed results were subsequently compared to determine the effect of incorporating fraud risk into the recommendation process.

3.4 Measures and evaluation metrics

The evaluation of the Risk-Aware Job Recommendation and Ranking component was designed to assess the system from two main perspectives: recommendation relevance and recommendation safety. Relevance measures were used to determine whether the generated Top-10 recommendation lists contained and appropriately ranked jobs

relevant to the corresponding test-user profiles. Safety measures were used to determine whether incorporating fraud-risk information reduced the risk associated with the recommended jobs. The same evaluation procedure was applied to both the relevance-based baseline and the proposed risk-aware approach to enable a direct comparison.

The evaluation used 50 test-user profiles, with each approach generating a Top-10 recommendation list for every profile. The relevance evaluation used the prepared ground-truth dataset, while the safety evaluation used the aggregated fraud-risk and safety scores associated with the recommended jobs. The selected measures were therefore intended to evaluate whether the proposed approach could improve recommendation safety while maintaining comparable recommendation relevance.

Recommendation Relevance Measures

Four ranking and retrieval measures were used to evaluate recommendation relevance: Precision@10, Average Precision at 10 (AP@10), Mean Reciprocal Rank at 10 (MRR@10), and Normalized Discounted Cumulative Gain at 10 (NDCG@10). These measures were calculated over the Top-10 recommendations generated for the test users. The actual evaluation output reports all four measures for both the baseline and proposed approaches.

- **Precision@10**

  Precision@10 measures the proportion of the first 10 recommended jobs that are relevant to the corresponding test-user profile. It is calculated as:

$$Precision@10 = \frac{\text{Number of relevant jobs in Top-10}}{10}$$

  A higher Precision@10 indicates that a greater proportion of the recommended jobs are relevant. Since the evaluation focuses specifically on the first 10 recommendations presented to a user, the @10 cutoff was used.

- **Average Precision@10 (AP@10)**

  AP@10 evaluates the ranking of relevant jobs within the Top-10 recommendation list by considering the precision achieved at the positions

where relevant jobs occur. It therefore considers not only how many relevant jobs are retrieved but also how they are distributed throughout the ranked list.

For a single user, AP@10 can be expressed as:

$$AP@10 = \frac{1}{R}\sum_{k=1}^{10} P@k \times rel_k$$

where R represents the number of relevant items considered for the evaluation, $P@k$ is the precision at position k, and rel_k indicates whether the item at position k is relevant.

The average AP@10 across the evaluated test users provides the reported overall measure. Average precision is designed to account for the positions of relevant results rather than treating all positions within the Top-10 equally.

- **Mean Reciprocal Rank@10 (MRR@10)**

  MRR@10 measures how highly the first relevant job appears within the Top-10 recommendations. For each test user, the reciprocal of the rank position of the first relevant recommendation is calculated:

$$MRR@10 = \frac{1}{N}\sum_{i=1}^{N} RR_i$$

  If no relevant job appears within the first 10 positions, the reciprocal-rank contribution is zero. MRR@10 is then calculated as the mean reciprocal rank across the evaluated users.

  A higher MRR@10 therefore indicates that users are more likely to encounter a relevant job near the beginning of the recommendation list.

- **Normalized Discounted Cumulative Gain@10 (NDCG@10)**

  NDCG@10 evaluates the quality of the ordering of the Top-10 recommendations by giving greater importance to relevant jobs appearing at higher positions. Unlike Precision@10, which does not distinguish between different positions within the Top-10 list, NDCG accounts for ranking position and can accommodate graded relevance judgements.

The discounted cumulative gain is calculated from the relevance values of the ranked results and is normalized against the ideal ranking:

$$NDCG@10 = \frac{DCG@10}{IDCG@10}$$

where *DCG@10* is the discounted cumulative gain of the generated ranking and *IDCG@10* is the discounted cumulative gain of the ideal ranking.

$$DCG@10 = \sum_{k=1}^{10} \frac{2^{rel_k} - 1}{\log_2(k+1)}$$

$$IDCG@10 = \sum_{k=1}^{10} \frac{2^{rel_k^*} - 1}{\log_2(k+1)}$$

An NDCG@10 value closer to 1 indicates that the relevant jobs are positioned closer to the ideal ordering. NDCG is particularly suitable for ranking evaluation because it considers both relevance and position.

Recommendation Safety Measures

The proposed component does not evaluate safety using only whether a job is labelled fraudulent or legitimate. Instead, the three fraud-related signals are first aggregated into a continuous Risk Score, which is converted into a corresponding Safety Score. The safety of the Top-10 recommendation lists is then evaluated using Average Risk@10, Average Safety@10, and Unsafe Rate@10.

The actual evaluation output reports all three safety measures for both the baseline and proposed approaches.

- **Average Risk@10**

  Average Risk@10 measures the mean aggregated fraud-risk score of the jobs appearing within the Top-10 recommendations.

  For a user *u*, the measure can be expressed as:

$$Average\ isk@10 = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{1}{10}\sum_{j=1}^{10} Risk_{ij}\right)$$

where $R_{u,j}$ is the aggregated fraud-risk score of the job at recommendation position j.

The overall Average Risk@10 is obtained by aggregating this measure across the evaluated test users.

A lower value indicates lower average fraud risk among the recommended jobs.

The risk score itself is produced from the three preceding fraud-detection outputs using the Entropy Weight Method. The implementation then retains both risk_score and safety_score for each job.

- **Average Safety@10**

  Average Safety@10 measures the mean safety score of the jobs appearing in the Top-10 recommendations.

  The proposed risk-aggregation implementation defines safety as the inverse of aggregated risk:

$$Average\ safety@10 = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{1}{10}\sum_{j=1}^{10} Safety_{ij}\right)$$

  Therefore:

$$Safety = 1 - Risk$$

  where $S_{u,j}$ is the safety score of the job at position j.

  A higher Average Safety@10 indicates a safer recommendation list. The implementation's unit tests also explicitly verify that the safety score is the inverse of the risk score.

- **Unsafe Rate@10**

  Unsafe Rate@10 measures the proportion of jobs within the Top-10 recommendations that meet the evaluation criterion for being considered unsafe.

  It can be expressed as:

$$Unsafe\ rate@10 = \frac{\text{Number of unsafe jobs in Top-10}}{\text{Total number of jobs in Top-10}} \times 100$$

A lower Unsafe Rate@10 indicates that fewer unsafe jobs are present among the recommendations.

3.5 Data analysis

The evaluation data was analysed by comparing the recommendation outputs generated by the relevance-based baseline and the proposed risk-aware approach. Both approaches were evaluated using the same test-user profiles and job dataset, allowing the effect of incorporating fraud-risk information into the ranking process to be examined under the same evaluation conditions.

The analysis was performed at both the overall evaluation level and the individual test-user level. For each test user, the Top-10 recommendation lists produced by the baseline and proposed approaches were compared. The comparison identified the number of jobs shared by the two recommendation lists, the jobs appearing only in the baseline list, and the jobs appearing only in the proposed list. The overlap percentage was calculated from the number of common jobs in the two Top-10 lists. This analysis was used to determine how substantially the introduction of safety-aware ranking changed the recommended job set.

The recommendation outputs were then analysed using the relevance and safety measures defined in Section 3.4. For recommendation relevance, the Precision@10, AP@10, MRR@10, and NDCG@10 values were compared between the baseline and proposed approaches. The analysis focused on whether the proposed approach maintained comparable relevance while incorporating an additional safety criterion into the ranking process.

For recommendation safety, Average Risk@10, Average Safety@10, and Unsafe Rate@10 were compared between the two approaches. Lower Average Risk@10 and Unsafe Rate@10 values indicate a reduction in the risk associated with the recommended jobs, while a higher Average Safety@10 indicates an improvement in the safety score of the recommendations. These measures were analysed together with

the relevance measures because an improvement in safety alone would not be sufficient if recommendation relevance were substantially reduced.

The analysis therefore used a comparative approach in which changes between the baseline and proposed results were examined rather than evaluating the proposed model in isolation. The resulting comparisons provide the basis for determining whether the proposed risk-aware ranking approach can incorporate fraud-risk information while maintaining recommendation relevance.

3.6 Ethics privacy security and responsible AI

This research was designed to avoid the collection and processing of personal information from real job seekers. The evaluation uses publicly available job-posting data and synthetic user profiles rather than real user accounts or personal employment records. Therefore, the evaluation does not require the collection of personally identifiable information from human participants. The use of synthetic profiles also allows different job-seeker scenarios to be represented without exposing the personal information of real individuals.

The job-posting data used in the experiment is treated as research data and is processed only for the purposes of recommendation and evaluation. The recommendation component does not independently collect additional personal information during the evaluation. The fraud-related indicators used by the component are supplied by the preceding fraud-detection components and are processed as model inputs for risk aggregation and ranking.

Responsible use of fraud-risk information is particularly important because the risk scores produced by the system are model-derived indicators rather than absolute evidence that a job or employer is fraudulent. Errors in the preceding fake-job, employer-legitimacy, or communication-scam components can therefore affect the risk and safety scores used by the recommendation component. Consequently, a high safety score should not be interpreted as a guarantee that a job opportunity is completely safe, and a higher risk score should be interpreted as an indication requiring caution rather than as definitive proof of fraud.

The recommendation process also involves a trade-off between job relevance and safety. The proposed ranking method therefore considers both Skill Match Score and Safety Score instead of removing jobs solely according to their risk values. This reduces the possibility of treating the fraud-risk indicators as the only decision criterion and keeps job suitability as part of the recommendation process. The TOPSIS implementation uses the two criteria with equal weights of 0.5 for skill relevance and 0.5 for safety.

The evaluation is conducted offline using synthetic users and a fixed job dataset. Therefore, the findings should not be interpreted as evidence of performance for all real-world job seekers or all forms of recruitment fraud. The use of synthetic profiles also means that the evaluation cannot fully represent the diversity, preferences, and behaviour of actual job seekers. These considerations are treated as limitations when interpreting the research findings.

3.7 Reproducibility and research artefacts

The research was implemented using a modular software structure in which the main processing stages are separated into skill matching, risk aggregation, and ranking modules. The implementation uses Python together with data-processing and machine-learning libraries. The research proposal identifies Python as the main implementation language, with scikit-learn used for TF-IDF and Cosine Similarity and Pandas and NumPy used for data processing and numerical computation.

The evaluation artefacts include the job-posting dataset, synthetic test-user profiles, ground-truth relevance data, baseline recommendation outputs, proposed recommendation outputs, and comparison outputs. The baseline evaluation script uses the same job dataset and test-user dataset to generate a Top-10 recommendation list for each test user, with the Top-K value explicitly configured as 10.

The implementation of the proposed ranking component is also maintained as a separate module. It combines the Skill Match Score and Safety Score into a two-criterion decision matrix, applies vector normalization and criterion weights, determines the ideal and anti-ideal solutions, calculates the distances from these solutions, and produces the TOPSIS closeness coefficient used to order the

recommendations. The implemented default weights are 0.5 for skill relevance and 0.5 for safety.

Software tests were used to verify important component behaviours. The risk aggregation tests verify that the expected output columns are produced, that safer jobs receive higher safety scores, that risk and safety are inversely related, and that the in-memory and CSV-based aggregation paths are consistent. The ranking tests verify that the TOPSIS score is produced and that, when skill-match scores are equal, a job with a higher safety score is ranked above a riskier job.

The evaluation process also stores intermediate and final outputs rather than relying only on manually observed results. The baseline and proposed recommendation files are subsequently compared on a per-user basis, and the comparison output records common jobs, baseline-only jobs, proposed-only jobs, and the corresponding overlap percentage. These artefacts provide traceability between the input data, recommendation-generation process, and evaluation results.

For reproducibility, the experimental configuration should be maintained together with the implementation, including the test-user dataset, Top-K value, ranking criterion weights, evaluation scripts, and generated result files. Maintaining these artefacts allows the evaluation procedure to be repeated using the same experimental configuration and provides a clear record of how the reported results were obtained.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The Risk-Aware Job Recommendation and Ranking component was designed to integrate job relevance and recruitment fraud risk into a single recommendation process. The requirements were derived from the identified research problem, the proposed conceptual framework, and the role of the component within the overall four-component system. The component is responsible for recommendation and ranking, while the underlying fraud-detection models are developed by the other components of the research project.

Functional requirements

- **FR1 – User profile input:**

  The component shall accept user information required for skill-based job matching, with the implemented evaluation using the skills associated with each synthetic test-user profile.

- **FR2 – Job-posting input:**

  The component shall accept the job-posting information required to calculate the relevance of available jobs to a user profile.

- **FR3 – Skill matching:**

  The component shall calculate a Skill Match Score between the user's skills and available job postings using TF-IDF representations and Cosine Similarity.

- **FR4 – Fraud-risk input:**

  The component shall receive fraud-risk information from the three preceding components: fake-job risk, employer-legitimacy risk, and communication-scam risk. These risk indicators are treated as inputs rather than being generated by the recommendation component itself. The overall architecture defines these three components as independent producers of risk information that is subsequently consumed by the recommendation component.

- **FR5 – Risk aggregation:**

The component shall combine the three fraud-risk indicators using the Entropy Weight Method to obtain an aggregated Risk Score.

- **FR6 – Safety calculation:**

  The component shall convert the aggregated Risk Score into a complementary Safety Score using the relationship:

  $$Safety\ Score = 1 - Risk\ Score$$

  A higher Safety Score therefore represents lower aggregated fraud risk.

- **FR7 – Multi-criteria ranking:**

  The component shall use the Skill Match Score and Safety Score as ranking criteria and apply TOPSIS to generate a risk-aware ranking.

- **FR8 – Recommendation generation**:

  The component shall produce a ranked list of job opportunities and provide the Top-10 recommendations used by the evaluation process.

- **FR9 – Baseline compatibility:**

  The system shall support a relevance-based baseline using Skill Match Scores so that the effect of incorporating fraud-risk information can be evaluated against a conventional relevance-oriented recommendation approach.

Non-functional requirements

- **NFR1 – Modularity:**

  The recommendation component should remain logically separated from the three fraud-detection components so that their models can be developed and updated independently while their risk outputs can still be consumed by the recommendation process. This modular interaction is part of the overall system architecture.

- **NFR2 – Maintainability:**

The skill matching, risk aggregation, and ranking stages should be separated into distinct processing modules so that changes to one stage do not require redesigning the complete recommendation process.

- **NFR3 – Reproducibility:**

  The component should use defined datasets, ranking parameters, and evaluation configurations so that the recommendation and evaluation processes can be repeated consistently.

- **NFR4 – Privacy:**

  The evaluation should avoid the use of personal information from real job seekers. Synthetic user profiles are therefore used for experimental evaluation.

- **NFR5 – Interpretability:**

  The ranking process should retain distinguishable relevance and safety criteria so that the effect of fraud risk on the final recommendation can be analysed.

4.2 Component architecture within the complete solution

The Risk-Aware Job Recommendation and Ranking component forms the fourth component of the overall four-component recruitment fraud detection and recommendation system. Its primary architectural role is to act as the integration point between the fraud-related outputs generated by the preceding components and the final job recommendation process.

The first three components analyse different sources of recruitment fraud. The Multilingual Fake Job Post Detection component produces a fake-job risk score, the Employer Legitimacy Verification component produces an employer-related risk score, and the Scam Communication Detection component produces a communication-scam risk score. These outputs are passed to the Risk-Aware Job Recommendation and Ranking component rather than being independently regenerated within it. This separation allows the fraud-detection components and recommendation component to operate as distinct functional units.

Within the fourth component, the architecture is divided into three processing stages: skill matching, fraud-risk aggregation, and risk-aware ranking. The first stage

processes the user skills and job information to calculate the Skill Match Score using TF-IDF and Cosine Similarity. In parallel, the fraud-related inputs from the preceding components are processed by the risk aggregation stage using the Entropy Weight Method. The resulting Risk Score is converted into a Safety Score.

The Skill Match Score and Safety Score are then provided to the final ranking stage. TOPSIS treats these values as the two decision criteria and produces a final ranking of the available jobs. The resulting Top-10 jobs form the recommendation output used by the evaluation process.

The architecture therefore separates fraud detection, risk aggregation, and recommendation ranking while maintaining a clear flow of information between them. This modular structure reflects the role of the individual component within the complete research system and allows the recommendation stage to incorporate outputs from multiple fraud-detection sources without directly implementing those detection models.

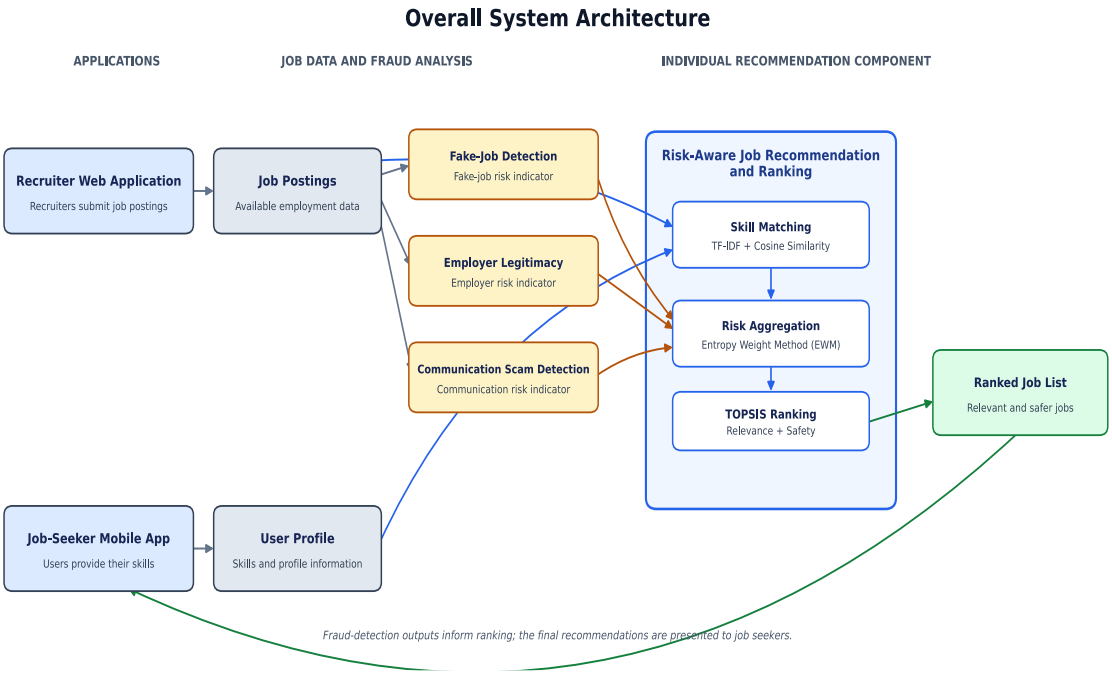


Figure 2: System Architecture

4.3 Design alternatives and decisions

Several design decisions were required to translate the conceptual framework into a practical recommendation architecture. The alternatives were considered according to the research objective, the role of the component within the overall system, and the requirement to incorporate multiple fraud-risk signals without replacing job relevance.

Skill matching approach

A content-based skill-matching approach using TF-IDF and Cosine Similarity was selected for the recommendation component. More complex semantic or deep-learning-based approaches could potentially capture richer relationships between textual descriptions; however, the selected design uses TF-IDF and Cosine Similarity because they directly support the content-based matching requirement established for the component and provide a transparent similarity score for each job. The selected approach also maintains consistency with the methodology defined for the research.

Fraud-risk aggregation approach

The three fraud-risk indicators could be combined using manually assigned or equal weights. However, such an approach would require predetermined assumptions about the relative importance of fake-job, employer, and communication risks. The Entropy Weight Method was therefore selected because it provides a data-driven mechanism for deriving the relative contribution of the three indicators from their information characteristics. The resulting weights are then used to obtain the aggregated Risk Score.

Risk representation

A binary safe/unsafe classification could have been used to remove jobs above a predefined risk level. However, the selected design represents risk continuously and converts it into a complementary Safety Score. This preserves differences between jobs with different levels of risk and allows safety to be incorporated as a ranking criterion rather than being used only as a filtering condition.

Ranking approach

A simple weighted scoring formula could combine relevance and safety directly. Instead, TOPSIS was selected because the recommendation problem contains multiple decision criteria and requires alternatives to be evaluated according to their distance from an ideal combination of high relevance and high safety. The implementation therefore uses Skill Match Score and Safety Score as the two ranking criteria.

Architectural decision

A modular architecture was selected instead of combining all fraud-detection and recommendation functionality into a single model. The overall system therefore allows the first three components to independently produce fraud-risk information while the recommendation component consumes these outputs. This separation supports independent development and evaluation of the fraud-detection components while maintaining their integration with the final recommendation process.

4.4 Detailed design

The detailed design of the Risk-Aware Job Recommendation and Ranking component follows the three-stage structure established in the conceptual framework: skill matching, fraud-risk aggregation, and risk-aware ranking. Each stage produces an intermediate output that is consumed by the subsequent stage.

Skill matching design

The first stage determines the relevance of each available job to the user's skills. The user skill information and job-related text are represented using TF-IDF. Cosine Similarity is then calculated between the user representation and the representation of each job. The resulting similarity value is used as the Skill Match Score.

The design can be represented as:

User Skills + Job Information → TF-IDF → Cosine Similarity → Skill Match Score

A higher Skill Match Score indicates stronger textual similarity between the user's skills and the corresponding job information. The resulting scores are retained for use by both the relevance-based baseline and the proposed risk-aware ranking process.

Fraud-risk aggregation design

The second stage receives three fraud-risk indicators from the preceding components of the overall system:

- Fake-job risk
- Employer-legitimacy risk
- Communication-scam risk

The three indicators are processed using the Entropy Weight Method. EWM determines the relative contribution of each risk criterion and produces weights that are used to calculate an aggregated Risk Score. The design therefore avoids treating the three fraud indicators as necessarily having identical importance.

The aggregated risk is then converted into a Safety Score:

$$Safety\ Score = 1 - Risk\ Score$$

The transformation allows the risk information to be represented in the same preference direction as the Skill Match Score, because higher values of both criteria are desirable during the final ranking process.

The processing flow is:

$$Fraud\ Risk\ Indicators \rightarrow EWM \rightarrow Risk\ Score \rightarrow Safety\ Score$$

Risk-aware ranking design

The final stage combines the Skill Match Score and Safety Score as the two decision criteria. The recommendation implementation constructs a decision matrix containing these two values for each job. The criteria are normalized before the TOPSIS procedure is applied.

The TOPSIS design consists of the following steps:

1. Construct the decision matrix using Skill Match Score and Safety Score.
2. Normalize the decision matrix.
3. Apply the criterion weights.
4. Determine the ideal best and ideal worst solutions.

5. Calculate the distance of each job from the ideal best and ideal worst solutions.

6. Calculate the TOPSIS closeness coefficient.

7. Sort the jobs according to the resulting TOPSIS score.

8. Select the highest-ranked jobs for the recommendation list.

In the implemented ranking module, the two criteria are assigned equal weights of 0.5 for Skill Match Score and 0.5 for Safety Score. The TOPSIS closeness coefficient is then used to order the jobs in descending order.

The final design can therefore be summarised as:

Skill Match Score + Safety Score → TOPSIS → Ranked Jobs → Top-10 Recommendations

This design ensures that the final recommendation is not determined solely by the similarity between the user and a job. Instead, the ranking considers both the suitability of the job for the user and the aggregated fraud-related safety information supplied by the overall system.

4.5 Standards and professional practice

The design of the recommendation component follows software engineering practices intended to support modularity, maintainability, testing, and responsible use of data. The component is divided into separate processing stages for skill matching, risk aggregation, and ranking. This separation of responsibilities reduces the dependency between individual processing stages and supports independent modification and testing.

The design also follows a consistent data-flow structure between the four components of the overall system. The three fraud-detection components provide risk information to the recommendation component using defined risk criteria, while the recommendation component is responsible for aggregating these inputs and incorporating them into the ranking process. This separation supports traceability of the source of each risk criterion and avoids duplicating fraud-detection functionality within the recommendation component.

Verification is supported through dedicated tests for important processing behaviour. The risk aggregation implementation includes tests for expected output fields, risk and safety relationships, and consistency of aggregation behaviour. The ranking implementation is also tested to verify the production of TOPSIS scores and the expected preference for safer jobs when the skill-match conditions are equivalent. These practices support confidence that the designed processing stages behave according to their intended functions.

The design also incorporates privacy-aware research practice. Real job-seeker personal information is not required for the experimental evaluation; synthetic user profiles are used instead. This reduces privacy risks associated with collecting and processing personal employment information.

Responsible AI considerations are incorporated into the interpretation of the ranking outputs. The Safety Score represents the aggregated risk information provided by the fraud-detection components and should not be interpreted as a guarantee that a job is safe. Similarly, the ranking system should not be interpreted as independently detecting every form of recruitment fraud. The design therefore preserves the distinction between model-derived risk information and verified real-world fraud.

Finally, the design is documented together with the datasets, processing stages, ranking configuration, evaluation procedures, and software tests. This supports traceability and reproducibility of the research artefact and allows the relationship between the conceptual framework, implementation, and evaluation to be maintained throughout the development process.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The Risk-Aware Job Recommendation and Ranking component was developed using Python as the primary programming language. Python was selected because it supports text processing, numerical computation, machine-learning operations, and the implementation of recommendation algorithms. The component follows a modular structure in which skill matching, fraud-risk aggregation, and final ranking are implemented as separate processing stages.

The implementation uses Pandas for loading, manipulating, and combining job-related data, while NumPy supports numerical operations required by the ranking and risk-calculation procedures. Scikit-learn is used for TF-IDF vectorisation and Cosine Similarity calculations. These libraries support the processing of job-seeker skills and job-related textual information to calculate relevance scores.

The recommendation pipeline is organised into separate modules for skill matching, risk aggregation, and TOPSIS-based ranking. This structure allows the individual stages to be tested independently and their outputs to be combined during recommendation generation. The risk aggregation stage uses the Entropy Weight Method to determine the relative contributions of the available fraud-risk indicators, while the ranking stage applies TOPSIS to evaluate jobs according to relevance and safety.

FastAPI is used to expose recommendation functionality through a backend API. Pydantic request models support the validation and structuring of incoming request data. The API includes a health-check endpoint and recommendation endpoints for processing recommendation requests. This provides an interface through which the recommendation component can communicate with other application components.

For experimental evaluation, the implementation also uses structured data files containing job records, risk indicators, synthetic user profiles, ground-truth relevance information, and generated recommendation results. This arrangement supports repeatable execution of the recommendation pipeline and comparison between the baseline and proposed approaches.

5.2 Component implementation

Skill Matching Module

The skill-matching module uses TF-IDF to represent job-seeker skills and job-related information as numerical vectors. Cosine Similarity is then calculated to produce a Skill Match Score for each job. In the intended application, user skills come from profiles created through the mobile application, while job information originates from recruiter-submitted postings.

Fraud-Risk Aggregation Module

The risk aggregation module is designed to combine three indicators: fake-job risk, employer-legitimacy risk, and communication-scam risk. The Entropy Weight Method (EWM) determines their relative weights and produces an aggregated Risk Score. The corresponding Safety Score is calculated as:

$$\text{Safety Score} = 1 - \text{Risk Score}$$

The module provides the risk and safety information required by the ranking stage.

TOPSIS-Based Ranking Module

The ranking module combines the Skill Match Score and Safety Score using TOPSIS. The decision matrix is normalised, criterion weights are applied, and the ideal best and worst solutions are determined. The algorithm calculates each job's closeness coefficient and sorts jobs in descending order to generate the ranked recommendations. The current implementation assigns equal weights of 0.5 to relevance and safety.

5.3 Integration with the other components

The Risk-Aware Job Recommendation and Ranking component forms the final stage of the overall four-component recruitment fraud detection and recommendation system. Recruiters submit job postings through the employer web application, while job seekers create profiles and provide their skills through the mobile application.

The three preceding components generate fraud-risk indicators for fake jobs, employer legitimacy, and scam communications. The recommendation component combines these indicators with the Skill Match Score to calculate an overall Risk Score and

Safety Score using the Entropy Weight Method. TOPSIS then ranks available jobs based on relevance and safety, producing a list of job recommendations for job seekers.

This modular architecture separates fraud detection from recommendation and ranking, allowing the outputs of the three detection components to influence the final recommendation process.

5.4 Verification and software testing

Unit tests were developed to verify the behaviour of the risk aggregation and ranking modules. The risk aggregation tests check the expected output structure, the inverse relationship between Risk Score and Safety Score, and consistency between CSV-based and in-memory processing.

The ranking tests verify TOPSIS score generation, descending ranking order, and whether a safer job receives a higher ranking when two jobs have equal skill-match scores. A pipeline integration test also checks the combined execution of skill matching, risk aggregation, and ranking when the required datasets are available.

These tests verify the main computational functions of the recommendation component. The quantitative performance of the proposed approach is evaluated separately in Chapter 6.

5.5 Deployment and operation

The recommendation component is implemented as a backend service that provides the functionality required for job matching, fraud-risk aggregation, and ranking. It processes job-seeker skills, available job information, and fraud-risk indicators to generate a ranked list of job recommendations. The generated results include job relevance and safety information to support the selection of suitable employment opportunities.

The component is designed to operate within the overall recruitment platform, connecting the job-seeker's profile and available job postings with the outputs of the fraud-detection components. This structure enables the recommendation process to consider both job suitability and recruitment safety when prioritising employment opportunities.

6 RESULTS AND VALIDATION

6.1 Research results

Recommendation Relevance Results

The Risk-Aware Job Recommendation and Ranking component was evaluated by comparing the proposed fraud-aware recommendation approach with a relevance-based baseline. The baseline used TF-IDF and Cosine Similarity to rank jobs according to skill relevance, while the proposed approach additionally incorporated the fraud-risk information generated by the preceding components and applied risk aggregation and TOPSIS-based ranking.

The evaluation used 50 synthetic test-user profiles representing five job domains: Information Technology, Business Development, Finance, Sales, and Human Resources, with 10 profiles representing each domain. The evaluation used a public job-posting dataset containing 1,167 job records and a ground-truth dataset containing 516 user-job records. For each test user, both approaches generated a Top-10 recommendation list, resulting in 500 recommendation records for each approach.

The evaluation was conducted from two main perspectives: recommendation relevance and recommendation safety. Relevance was assessed using Precision@10, AP@10, MRR@10, and NDCG@10, while safety was assessed using Average Risk@10, Average Safety@10, and Unsafe Rate@10. The same test users and job data were used for both approaches to enable direct comparison.

Table 1: Baseline and Proposed Recommendation Relevance Evaluation Results

Metric	Baseline	Proposed	Change
Precision@10	0.6160	0.6160	0
AP@10	0.8612	0.8627	+0.0015
MRR@10	1.0000	1.0000	0
NDCG@10	0.9648	0.9655	+0.0007

The Precision@10 value was 0.6160 for both approaches, indicating that the incorporation of fraud-risk information did not change the measured proportion of relevant jobs within the Top-10 recommendations.

The AP@10 value increased slightly from 0.8612 to 0.8627, representing an improvement of 0.0015. This indicates a small improvement in the ordering of relevant jobs within the evaluated recommendation lists.

The MRR@10 value remained unchanged at 1.0000 for both approaches. Therefore, the introduction of fraud awareness did not change the measured reciprocal rank of the first relevant recommendation.

NDCG@10 increased from 0.9648 for the baseline to 0.9655 for the proposed approach, representing an improvement of 0.0007. This indicates that the proposed ranking maintained, and slightly improved, the position of relevant jobs within the Top-10 recommendations.

Overall, the relevance results show that incorporating fraud-risk information did not reduce the measured recommendation relevance. Precision@10 and MRR@10 remained unchanged, while AP@10 and NDCG@10 showed small positive changes. The complete metric values are also reported in the evaluation comparison table.

Recommendation Safety Results

Table 2: Recommendation Safety Results

Metric	Baseline	Proposed	Change
Average Risk@10	0.1006	0.0900	−0.0106
Average Safety@10	0.8994	0.9100	+0.0106
Unsafe Rate@10	2.80%	2.40%	−0.40%

The proposed approach reduced the Average Risk@10 from 0.1006 to 0.0900, corresponding to an absolute reduction of 0.0106. This represents an approximate 10.5% reduction relative to the baseline risk.

The Average Safety@10 increased from 0.8994 to 0.9100, representing an improvement of 0.0106, or approximately 1.18% relative to the baseline safety value.

The Unsafe Rate@10 decreased from 2.80% to 2.40%, corresponding to a reduction of 0.40 percentage points. In relative terms, this represents approximately a 14.29% reduction in the unsafe recommendation rate.

These results indicate that the proposed risk-aware ranking approach produced recommendation lists with lower aggregated fraud risk and higher safety than the relevance-based baseline. The evaluation comparison table confirms the same safety results.

Overall Baseline–Proposed Comparison

The combined results demonstrate a consistent difference between the two approaches. The proposed approach produced lower recommendation risk and a lower unsafe rate while maintaining the same Precision@10 and MRR@10 values as the baseline. It also produced small improvements in AP@10 and NDCG@10.

The results can therefore be summarised as follows:

- Precision@10: unchanged.
- AP@10: increased by 0.0015.
- MRR@10: unchanged.
- NDCG@10: increased by 0.0007.
- Average Risk@10: decreased by 0.0106.
- Average Safety@10: increased by 0.0106.
- Unsafe Rate@10: decreased by 0.40 percentage points.

The proposed approach therefore achieved the primary evaluation objective of improving the safety characteristics of the recommendations without an observed reduction in the measured relevance metrics.

This result is important in relation to the research question because the purpose of introducing fraud awareness was not simply to remove risky jobs, but to incorporate safety into the ranking while preserving the ability of the system to recommend relevant employment opportunities.

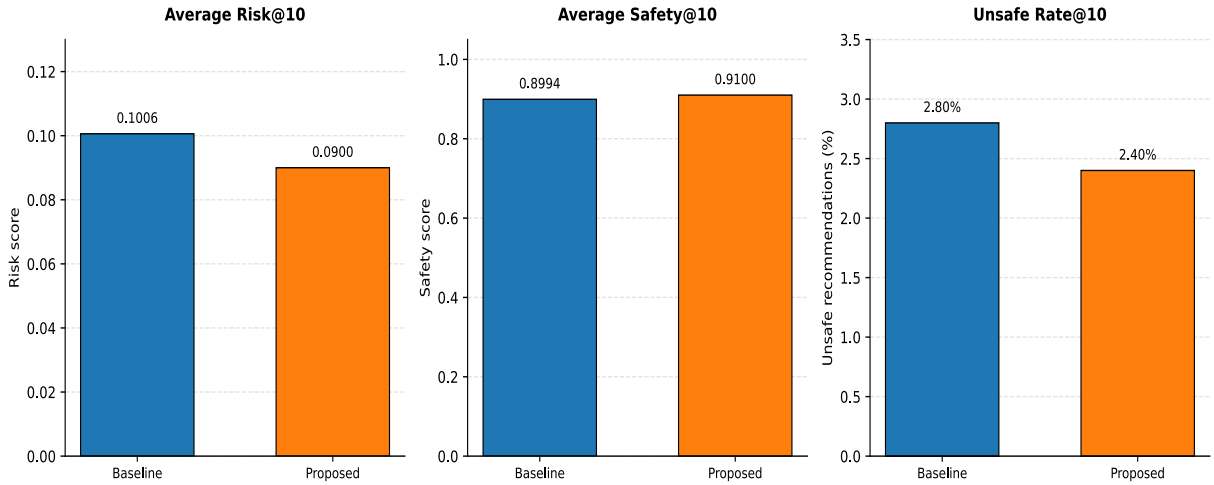


Figure 3: Research Results

6.2 Product and technical results

In addition to the quantitative evaluation, the development of the component produced a functional recommendation and ranking pipeline consisting of skill matching, fraud-risk aggregation, safety calculation, and multi-criteria ranking.

The skill-matching stage successfully generated a Skill Match Score for the available job opportunities based on the similarity between the job-seeker's skills and job information used by the recommendation pipeline. These scores provide the relevance criterion used by both the baseline and proposed approaches.

The risk aggregation stage successfully incorporated the three fraud-related indicators supplied by the preceding components. The Entropy Weight Method was used to determine the relative contribution of the risk indicators before calculating the aggregated Risk Score and corresponding Safety Score.

The evaluation produced the following EWM weights for the three risk indicators:

Table 3: EWM weights for the three risk indicators

Risk Indicator	EWM Weight
Fake-job risk	0.2499
Employer-legitimacy risk	0.2982
Communication-scam risk	0.4519

The communication-scam indicator received the highest weight in the evaluated data, followed by employer-legitimacy risk and fake-job risk. These weights represent the relative contribution determined by the EWM procedure for the evaluation data rather than manually assigned preferences.

The resulting Skill Match Score and Safety Score were then supplied to the TOPSIS ranking stage. The proposed approach generated a Top-10 list for each of the 50 test users, allowing the final recommendations to be evaluated against the baseline.

The component therefore produced the intended technical outputs required for the risk-aware recommendation process: skill relevance, aggregated fraud risk, safety, and final ranking.

6.3 User or stakeholder validation

No formal validation involving real job seekers, recruiters, or other human participants was conducted as part of this individual research evaluation. The study therefore does not make claims about user satisfaction, perceived usefulness, usability, or behavioural changes resulting from the recommendations.

Instead, validation was performed through quantitative experimentation using synthetic test-user profiles, the public job-posting dataset, prepared ground-truth relevance information, and fraud-risk information supplied by the preceding components. This approach allowed the proposed recommendation method to be evaluated under controlled conditions without collecting personal information from real job seekers.

The absence of human-participant validation means that the results demonstrate the technical and experimental performance of the recommendation component rather than its effectiveness from a real user's perspective. User-centred validation could be considered as future work after the complete integrated application is finalized.

6.4 Integration results

The Risk-Aware Job Recommendation and Ranking component was evaluated as the downstream ranking stage of the overall four-component framework. The proposed

approach receives the fraud-related outputs of the preceding components and combines them with job relevance during recommendation generation.

The integration of the risk information resulted in changes to the recommendation lists while retaining a high degree of similarity with the baseline results. The baseline and proposed Top-10 recommendation lists achieved an average overlap of 96.80% across the evaluated users. The comparison was calculated by determining the proportion of common jobs between the two Top-10 lists for each user.

The high overlap indicates that incorporating fraud risk did not completely replace the relevance-based recommendations. Instead, the proposed ranking modified the recommendations sufficiently to improve their safety characteristics while retaining most of the jobs selected by the relevance-based baseline.

This provides an important integration result for the proposed framework. The fraud-detection outputs generated by the preceding components can be incorporated into the downstream recommendation process without causing a large change in the set of jobs presented to users.

6.5 Summary of findings

The evaluation results provide evidence that the proposed Risk-Aware Job Recommendation and Ranking approach can incorporate recruitment-fraud information into job ranking while maintaining the measured recommendation relevance.

The proposed approach achieved the same Precision@10 and MRR@10 as the baseline, while AP@10 and NDCG@10 showed small improvements. At the same time, Average Risk@10 decreased from 0.1006 to 0.0900, Average Safety@10 increased from 0.8994 to 0.9100, and Unsafe Rate@10 decreased from 2.80% to 2.40%.

The average Top-10 overlap of 96.80% further indicates that the proposed approach retained most of the jobs identified by the relevance-based baseline while making ranking changes based on safety.

Therefore, within the conditions of the conducted experiment, the results support the feasibility of integrating multiple fraud-risk indicators into a job recommendation and ranking process without an observed reduction in recommendation relevance and with measurable improvements in the selected safety measures.

However, these results should be interpreted as experimental evidence from the selected dataset and synthetic user profiles rather than as proof that the approach will perform identically with real job seekers and live recruitment data.

7 DISCUSSION

7.1 Interpretation of findings

The results indicate that incorporating recruitment fraud risk into the job recommendation process can improve the safety characteristics of recommended jobs while maintaining the measured recommendation relevance. The proposed approach achieved the same Precision@10 and MRR@10 as the relevance-based baseline, while AP@10 increased from 0.8612 to 0.8627 and NDCG@10 increased from 0.9648 to 0.9655. At the same time, Average Risk@10 decreased from 0.1006 to 0.0900, Average Safety@10 increased from 0.8994 to 0.9100, and Unsafe Rate@10 decreased from 2.80% to 2.40%.

The most important finding is therefore not a large improvement in recommendation relevance, but the ability to introduce an additional safety consideration without an observed reduction in relevance. This is consistent with the purpose of the proposed component, which is to balance job suitability with recruitment-related safety rather than to replace relevance-based matching entirely.

The small changes in the relevance metrics can also be understood in the context of the recommendation lists generated by the two approaches. The baseline and proposed Top-10 lists had an average overlap of 96.80%. This indicates that most of the jobs selected by the relevance-based baseline remained present in the proposed recommendations. The proposed method therefore did not completely change the recommendation set; instead, the inclusion of safety as an additional ranking criterion resulted in relatively limited changes to the selected jobs.

The safety results show a clearer difference between the two approaches. The reduction in Average Risk@10 from 0.1006 to 0.0900 indicates that the proposed ranking placed recommendations with lower aggregated fraud risk higher within the evaluated Top-10 results. The corresponding increase in Average Safety@10 and reduction in Unsafe Rate@10 provide additional evidence that the changes in ranking were directed towards safer recommendations.

The results therefore support the underlying design principle of the proposed approach: fraud information can be treated as an additional decision criterion rather than as a

separate detection result that has no influence on recommendation ranking. The three fraud-related indicators are first aggregated into a common risk representation and then converted into a safety criterion that can be considered alongside job relevance.

However, the results should not be interpreted as evidence that the proposed method completely eliminates recruitment fraud. The evaluation demonstrates a reduction in the measured risk of the recommendations under the selected experimental conditions. The effectiveness of the final recommendations remains dependent on the quality of the fraud-risk information supplied by the preceding components and the characteristics of the evaluation dataset.

Overall, the findings provide experimental support for the research question. Within the evaluated setting, fraud risk was incorporated into the recommendation process without an observed reduction in the selected relevance measures, while the selected safety measures showed improvement.

7.2 Comparison with literature baselines and existing solutions

The findings can be interpreted in relation to the four studies selected in the literature review. These studies address different parts of the broader problem, and therefore direct numerical comparison with their reported results is not appropriate because their **datasets, experimental settings, objectives, and evaluation measures differ.**

Comparison with Fraudulent Job Detection

Chiraratanasopha and Chay-intr (2022) investigated fraudulent job recruitment using fraud-oriented features related to missing information, exaggeration, and credibility. Their best reported model achieved 97.64% accuracy, with precision of 0.97 and recall of 0.99.

Their work demonstrates that fraud-related characteristics can be used effectively to distinguish fraudulent and legitimate job advertisements. However, the purpose of their system is fraud classification rather than recommendation and ranking. The detection result is therefore produced before the recommendation decision rather than being used to determine which jobs should be prioritised for a particular user.

The proposed research extends this general direction by treating fraud information as an input to the recommendation process. Rather than asking only whether an advertisement is fraudulent, the proposed component considers the available fraud-risk information together with the relevance of the job to the user. This distinction is important because a recommendation system must ultimately decide not only whether an opportunity presents risk, but also how that opportunity should be positioned relative to other available opportunities.

Comparison with Content-Based Job Recommendation

Winardi et al. (2025) developed a content-based job recommendation system using TF-IDF and Cosine Similarity. Their system considered attributes including salary, education, skills, interests, and job location and reported a MAP score of 0.798.

This study is particularly relevant to the relevance component of the proposed research because both approaches use TF-IDF and Cosine Similarity to establish similarity between user-related information and job information.

However, the primary objective of the approach proposed by Winardi et al. is matching users with suitable employment opportunities. Recruitment fraud and employer trustworthiness are not incorporated into the recommendation criteria. The proposed research retains the content-based relevance approach but introduces a second decision dimension representing recruitment safety.

Consequently, the contribution of the proposed component is not to replace content-based job matching, but to extend the decision process beyond relevance alone.

Comparison with Risk-Aware Job Recommendation

Honmane et al. (2026) demonstrate another important direction by combining risk assessment with job recommendation. Their framework predicts an individual's risk of job loss using employment-related information and machine-learning models including Random Forest, XGBoost, and Logistic Regression, followed by job matching and career recommendations.

The similarity between the two approaches is the principle of incorporating risk information into a recommendation-oriented system. However, the meaning of risk

differs substantially. Honmane et al. consider the risk of job loss faced by the individual, whereas the present research considers risk associated with the recruitment opportunity, including fake job, employer-legitimacy, and communication-scam indicators.

Therefore, the two approaches demonstrate complementary uses of risk-aware recommendation. The former considers whether a person's employment situation may be unstable, while the proposed component considers whether an employment opportunity itself presents potential recruitment-related risk.

Comparison with Multi-Criteria Recommendation

Canay's 2026 CF-RL-TOPSIS framework is relevant to the proposed research because it demonstrates the use of entropy-weighted TOPSIS within an interpretable multi-criteria recommendation framework. The study evaluates technology talent recommendation using multiple occupation-related criteria and applies ranking evaluation including NDCG.

The proposed research shares the use of entropy-based weighting and TOPSIS for multi-criteria recommendation. However, the decision criteria differ. Canay's framework focuses on talent and career progression characteristics, whereas the proposed component uses job relevance and recruitment safety as the principal ranking criteria.

Furthermore, Canay's work combines collaborative filtering, a reinforcement-style branch, and an entropy-weighted TOPSIS branch. The proposed research uses a simpler architecture focused specifically on integrating fraud-risk information into job recommendation.

The selected study is also treated as a 2026 preprint in this research rather than as an established journal publication. Therefore, it is used primarily as methodological evidence for the applicability of entropy weighting and TOPSIS to recommendation ranking.

Overall Comparison

The comparison demonstrates that the proposed research is positioned at the intersection of several existing research directions. Fraud detection provides mechanisms for generating fraud-related evidence, content-based recommendation provides mechanisms for measuring job relevance, risk-aware recommendation demonstrates the value of incorporating risk into recommendation, and multi-criteria decision-making provides mechanisms for combining several decision criteria.

The contribution of the proposed component lies in connecting these aspects within the specific context of recruitment-fraud-aware job recommendation. The selected literature addresses these aspects largely as separate problems, whereas the proposed approach uses fraud-risk information as part of the final recommendation decision.

7.3 Research contribution

The primary research contribution is the development and evaluation of a risk-aware job recommendation and ranking approach that incorporates recruitment fraud risk into the recommendation decision.

First, the research contributes an integration mechanism for multiple recruitment-fraud indicators. Fake-job risk, employer-legitimacy risk, and communication-scam risk are treated as separate inputs and aggregated into a common risk representation using the Entropy Weight Method. This allows information originating from different fraud-detection components to influence the final recommendation ranking.

Second, the research contributes a multi-criteria recommendation strategy that considers both job relevance and safety. Instead of treating fraud detection as a separate classification task, the proposed approach converts the aggregated risk into a Safety Score and incorporates it alongside Skill Match Score during TOPSIS ranking.

Third, the research provides empirical evidence concerning the trade-off between relevance and safety. In the conducted evaluation, the proposed approach maintained Precision@10 and MRR@10 while producing small improvements in AP@10 and NDCG@10. At the same time, Average Risk@10 and Unsafe Rate@10 decreased. This provides evidence that fraud awareness can be introduced into the ranking process without an observed reduction in the measured relevance of the recommendations.

Finally, the research contributes an evaluation perspective in which recommendation systems are assessed using both relevance and safety rather than recommendation relevance alone. This is important for recruitment applications because a highly relevant recommendation may still be undesirable if the associated opportunity presents substantial fraud risk.

7.4 Product contribution

From a product perspective, the developed component provides the final decision-making layer that connects fraud analysis with personalized job recommendation within the overall system.

The first three components of the overall framework produce fraud-related information concerning fake jobs, employers, and scam communications. The Risk-Aware Job Recommendation and Ranking component consumes these outputs and transforms them into a form that can directly influence the jobs presented to a user.

The component therefore contributes four main product capabilities. First, it provides skill-based job matching. Second, it combines multiple fraud-risk indicators into an overall risk and safety representation. Third, it uses both relevance and safety during ranking. Finally, it produces a Top-10 ranked recommendation list that can be delivered to the application layer.

This makes the component an important integration point within the overall product. Without the recommendation layer, the fraud-detection outputs would primarily indicate that risk exists. The proposed component allows those outputs to influence the ordering and selection of job opportunities presented to the user.

The evaluation results further indicate that this integration can be achieved while retaining a high degree of similarity with the relevance-based recommendations, as demonstrated by the 96.80% average Top-10 overlap.

7.5 Limitations and threats to validity

Several limitations should be considered when interpreting the findings.

Dataset limitations

The evaluation was conducted using a public job-posting dataset containing 1,167 job records. Although this provides a consistent experimental environment, a single public dataset cannot represent the full diversity of real-world online recruitment platforms. Job descriptions, skill requirements, employers, industries, geographical regions, and fraud patterns may differ considerably across platforms.

Therefore, the reported performance should not be interpreted as evidence that the same numerical improvements would necessarily occur on other datasets.

Synthetic user profiles

The evaluation used 50 synthetic test-user profiles rather than real job seekers. Synthetic profiles provide a controlled way to evaluate different skill scenarios without collecting personal information, but they cannot fully represent the complexity of real users' preferences, career goals, experience, or decision-making behaviour.

Consequently, the evaluation demonstrates algorithmic performance under simulated user conditions rather than real-world user satisfaction or usefulness.

Ground-truth limitations

The relevance evaluation depends on the prepared ground-truth user-job relationships. The quality of the relevance metrics is therefore influenced by how accurately the ground-truth information represents appropriate job matches.

The evaluation should consequently be interpreted as performance against the prepared relevance reference rather than an absolute measurement of whether every recommendation would be considered suitable by a real job seeker.

Dependence on preceding fraud-detection components

The proposed component does not independently generate the fake-job, employer-legitimacy, or communication-scam predictions. Instead, it consumes the outputs of the preceding components.

Therefore, errors in those upstream predictions can propagate into the risk aggregation and final ranking stages. An incorrect fraud-risk input may cause an otherwise relevant job to receive an inappropriate safety score.

Limitations of TF-IDF-based matching

The skill-matching stage relies on TF-IDF and Cosine Similarity. These methods are useful for lexical similarity but may not fully capture semantic relationships between different terms that express similar skills.

For example, two skills may be conceptually related but represented using different terminology. A purely lexical similarity approach may not identify such relationships as effectively as more semantically oriented approaches.

This limitation is particularly relevant because the current component prioritizes a relatively simple and interpretable matching strategy rather than using more complex semantic models.

Dependence of EWM on the evaluation data

The EWM weights are derived from the information distribution of the evaluated risk indicators. Therefore, the resulting weights are data-dependent rather than universal weights for recruitment fraud.

The weights obtained in this evaluation—approximately 0.2499 for fake-job risk, 0.2982 for employer-legitimacy risk, and 0.4519 for communication-scam risk—should therefore not be interpreted as generally applicable importance values for all recruitment environments.

TOPSIS weighting assumption

The current ranking implementation gives equal importance to Skill Match Score and Safety Score. This 0.5/0.5 weighting provides a straightforward and transparent evaluation configuration, but it represents a design assumption.

Different users, platforms, or application contexts may prefer different relevance–safety trade-offs. Therefore, further research could investigate alternative weighting strategies and sensitivity to changes in these criterion weights.

Limited evaluation scope

The reported results focus on Top-10 recommendations. Performance at other values of K may differ. Similarly, the experiment does not establish how the approach behaves under continuously changing job inventories or changing fraud patterns.

Lack of real-world user validation

No formal user study was conducted. Therefore, conclusions about usability, user trust, satisfaction, or willingness to use fraud-aware recommendations cannot be drawn from the current evaluation.

Deployment limitations

The evaluation demonstrates the functional recommendation pipeline and its integration with the overall framework, but it should not be interpreted as evidence of production-scale deployment. Further testing would be required for factors such as scalability, latency under large job inventories, continuously updated fraud signals, and operational reliability.

Taken together, these limitations indicate that the findings provide controlled experimental evidence of the proposed approach, rather than a guarantee of performance across all real-world recruitment environments.

7.6 Implications

The findings have implications for both research and the development of recruitment recommendation systems.

Research implications

The results suggest that recommendation relevance and recruitment safety can be treated as complementary decision criteria rather than as completely separate research problems. The observed reduction in risk without an observed reduction in relevance supports further investigation into multi-criteria approaches for fraud-aware recommendation.

The research also demonstrates that fraud-detection outputs can have value beyond classification. Once converted into suitable risk indicators, they can be incorporated into downstream decision-making processes such as recommendation and ranking.

Product implications

For job platforms, the findings suggest that fraud-related information could potentially be incorporated into recommendation ranking rather than being displayed only as a separate warning or detection result. A recommendation system could therefore consider both whether a job matches a user's skills and whether the available evidence indicates potential recruitment risk.

However, such a system should present risk information as an indication rather than a guarantee. A risk-aware recommendation system should not imply that a high-safety score proves that a job or employer is completely legitimate.

Job-seeker implications

From a job-seeker perspective, the proposed approach supports a recommendation process in which relevant opportunities are considered together with available safety information. This can potentially help users prioritise opportunities that provide a stronger balance between suitability and lower observed recruitment risk.

The approach should nevertheless be treated as a decision-support mechanism rather than a replacement for user judgement or independent verification of employment opportunities.

Future research implications

The current findings provide several directions for future research. Larger and more diverse job datasets could be used to examine generalisation across industries and recruitment platforms. Real-user studies could evaluate whether risk-aware ranking improves perceived trust and decision quality. More semantically expressive skill-matching techniques could also be investigated to address the limitations of lexical similarity.

Further research could additionally investigate dynamic criterion weighting, alternative multi-criteria ranking methods, and sensitivity analysis of the relevance–safety trade-off. These extensions could determine whether the observed safety improvements remain consistent under different datasets, user profiles, risk distributions, and ranking configurations.

Overall, the findings support the broader implication that job recommendation systems can be designed not only to identify suitable opportunities, but also to consider the safety of those opportunities as part of the recommendation decision.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research addressed the problem of incorporating recruitment fraud risk into job recommendation and ranking while maintaining the relevance of recommended employment opportunities. The research focused on the Risk-Aware Job Recommendation and Ranking component within the overall four-component recruitment fraud detection and recommendation framework.

The proposed approach combines TF-IDF and Cosine Similarity to determine the relevance between a job seeker's skills and available job opportunities. Fraud-related information obtained from the fake-job detection, employer-legitimacy verification, and communication-scam detection components is then aggregated using the Entropy Weight Method (EWM). The resulting Risk Score is converted into a Safety Score, which is subsequently considered together with the Skill Match Score during TOPSIS-based multi-criteria ranking.

The developed component was evaluated by comparing the proposed risk-aware approach with a relevance-based baseline. The evaluation results showed that the proposed approach maintained the measured recommendation relevance while improving the safety characteristics of the recommended jobs. Precision@10 and MRR@10 remained unchanged, while AP@10 and NDCG@10 showed small improvements. At the same time, Average Risk@10 decreased from 0.1006 to 0.0900, Average Safety@10 increased from 0.8994 to 0.9100, and Unsafe Rate@10 decreased from 2.80% to 2.40%.

The findings therefore provide experimental evidence that fraud-risk information can be incorporated into the recommendation and ranking process without an observed reduction in the selected relevance measures. The high average Top-10 overlap of 96.80% further indicates that the proposed approach retained most of the recommendations produced by the relevance-based baseline while introducing safety considerations into their ranking.

Overall, the research demonstrates the feasibility of combining job relevance and recruitment safety within a unified recommendation process. The developed

component provides a link between the fraud-analysis outputs of the overall framework and the final job recommendations presented to job seekers. Although the evaluation was conducted using a public job dataset and synthetic user profiles, the results provide a foundation for further development and validation using real-world recruitment data and users.

8.2 Achievement of objectives

The research objectives established at the beginning of the study were addressed through the design, implementation, and evaluation of the Risk-Aware Job Recommendation and Ranking component.

Objective 1: To develop a skill-matching mechanism using TF-IDF and Cosine Similarity.

This objective was achieved through the implementation of the skill-matching stage. TF-IDF was used to represent the job seeker's skills and job-related information, while Cosine Similarity was used to calculate the Skill Match Score for available job opportunities.

Objective 2: To incorporate multiple fraud-risk indicators into the recommendation process.

This objective was achieved by incorporating three fraud-related indicators obtained from the preceding components of the overall framework: fake-job risk, employer-legitimacy risk, and communication-scam risk.

Objective 3: To aggregate the fraud-risk indicators using the Entropy Weight Method.

This objective was achieved through the risk aggregation stage. EWM was applied to determine the relative weights of the three risk indicators and combine them into an overall Risk Score.

Objective 4: To convert the aggregated risk into a Safety Score.

This objective was achieved by transforming the aggregated Risk Score into its complementary Safety Score. This enabled lower-risk jobs to receive higher safety values for use during the final ranking process.

Objective 5: To apply TOPSIS to rank jobs according to relevance and safety.

This objective was achieved by using the Skill Match Score and Safety Score as the two criteria in the TOPSIS ranking process. The approach produced a final ranking based on the relative closeness of each job to the desired combination of high relevance and high safety.

Objective 6: To generate job recommendations that balance relevance and safety.

This objective was supported by the experimental results. Compared with the relevance-based baseline, the proposed approach maintained the measured relevance performance while reducing Average Risk@10 and Unsafe Rate@10 and increasing Average Safety@10. Therefore, the results demonstrate that the proposed approach can balance job relevance with recruitment safety under the conditions of the evaluation.

Overall, the objectives were addressed through the development and evaluation of the proposed recommendation and ranking component.

8.3 Individual research and product contributions

The individual research contribution of this study is the design and evaluation of a Risk-Aware Job Recommendation and Ranking approach that incorporates recruitment-fraud information into the final recommendation decision. The research combines skill-based relevance measurement with multiple fraud-risk indicators, allowing job recommendations to be assessed using both suitability and safety.

The research contributes a structured three-stage approach consisting of skill matching, fraud-risk aggregation, and multi-criteria ranking. TF-IDF and Cosine Similarity are used to determine job relevance, EWM is used to aggregate the fraud-risk indicators, and TOPSIS is used to rank jobs using relevance and safety. The evaluation further provides evidence that this integration can reduce the measured risk of the Top-10 recommendations without an observed reduction in the selected relevance measures.

From a product perspective, the developed component provides the recommendation and ranking layer within the overall recruitment platform. It connects the job-seeker's

profile and skills with available job opportunities and consumes fraud-related outputs from the preceding research components. The component transforms these inputs into ranked job recommendations that consider both skill relevance and recruitment safety.

The component therefore provides an important link between fraud detection and actionable job recommendation. Instead of leaving fraud-related outputs as independent detection results, the information can influence which opportunities are prioritised for the job seeker.

8.4 Future directions

Several directions can be considered to further develop and evaluate the proposed Risk-Aware Job Recommendation and Ranking approach.

First, the evaluation could be extended using larger and more diverse job datasets covering additional industries, job roles, and recruitment environments. This would provide a stronger basis for assessing the generalisability of the proposed approach beyond the dataset used in the current study.

Second, future work could incorporate real job-seeker profiles and user-centred evaluation. Real users could provide feedback on recommendation relevance, perceived safety, usability, and trust. This would complement the current quantitative evaluation based on synthetic test-user profiles.

Third, the skill-matching stage could be extended with more semantically expressive techniques. Such approaches could help identify relationships between skills that have similar meanings but different textual representations, addressing limitations associated with lexical similarity.

Fourth, future research could investigate alternative or adaptive criterion-weighting strategies for the TOPSIS stage. The current approach gives equal importance to relevance and safety. Future work could examine whether these weights should vary according to user preferences, platform policies, or different recruitment contexts.

Fifth, the fraud-risk information could be updated dynamically as new evidence becomes available. This could allow recommendation rankings to respond to changes in job risk rather than relying on static risk information.

Finally, the complete integrated application could be evaluated under real-world operational conditions, including larger numbers of job postings, concurrent users, response time, scalability, and continuous interaction between the employer web application, job-seeker mobile application, fraud-detection components, and recommendation service.

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GLOSSARY

Content-Based Recommendation – A recommendation approach that identifies relevant items by comparing their characteristics with a user's preferences or profile.

Cosine Similarity – A mathematical measure used to determine how similar two vectors are by comparing the angle between them.

Entropy Weight Method – A method for assigning weights to evaluation criteria based on the amount of information or variation present in the data.

Fraud Risk Score – A numerical value representing the estimated level of fraud risk associated with a job posting, employer, or communication.

Ground Truth – Reference labels or values used to evaluate the performance of a system against expected results.

Job Recommendation – The process of identifying and presenting job opportunities that match a job seeker's profile, skills, or preferences.

Multi-Criteria Decision-Making – A decision-making approach that evaluates alternatives using multiple criteria.

Risk-Aware Ranking – A ranking approach that considers both the relevance of an item and its associated risk when determining its position in a recommendation list.

Safety Score – A numerical value representing the estimated safety of a job opportunity, derived from its aggregated fraud risk score.

Skill Matching – The process of comparing a job seeker's skills with the skills or requirements specified in a job description.

TF-IDF – A text-weighting technique that represents the importance of words in a document relative to a collection of documents.

Top-K Recommendations – A fixed number of the highest-ranked items returned by a recommendation system, where K represents the number of items selected.

TOPSIS – A multi-criteria ranking method that evaluates alternatives according to their distance from an ideal solution and a negative-ideal solution.

APPENDICES

Appendix A Research instruments or extended results

A.1 Evaluation Configuration

Summarise the evaluation setup, including the 1,167 job records, 50 synthetic user profiles, 516 user–job ground-truth records, and Top-10 recommendation setting.

A.2 Detailed Relevance Evaluation

Provide a table comparing the baseline and proposed approaches using Precision@10, AP@10, MRR@10, and NDCG@10.

A.3 Detailed Safety Evaluation

Provide a table containing Average Risk@10, Average Safety@10, and Unsafe Rate@10 for both approaches.

A.4 Entropy Weight Method Results

Include the calculated weights for the three fraud-risk indicators:

- Fake-job risk: 0.2499
- Employer-legitimacy risk: 0.2982
- Communication-scam risk: 0.4519

A.5 Additional Supporting Evidence

Include representative recommendation outputs, additional evaluation tables, or selected test cases if they help demonstrate how the system works.

Appendix B Academic integrity and AI use disclosure

B.1 AI Tools Used

List the AI tools actually used during the preparation of this thesis and the associated research work.

B.2 Purpose of Use

Describe the specific purposes for which each tool was used, such as understanding technical concepts, improving language clarity, assisting with Python figure-generation code, or organising thesis content, where applicable.

B.3 Representative Prompts and Workflows

Provide representative examples of prompts or workflows used, where required by the applicable academic policy.

B.4 Verification and Author Responsibility

Explain how AI-assisted outputs were checked against the research methodology, implementation results, original sources, and relevant academic references. Confirm that the final content and reported results were reviewed by the author.

B.5 Affected Sections and Artefacts

Identify the thesis sections, figures, code, or other artefacts for which AI assistance was used, as applicable.